Cramer's Rule also provides another strategy for finding A^{-1}. That is, we want to solve

$$AA^{-1} = I$$

We will show how to find one specific entry of A^{-1}
To that end, focus on the first column of A^{-1}, say that is given by

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

so Ax is to be the first column of I, i.e. we need

$$Ax = e_1$$

Let us attempt to find the third entry of x, namely x_3.

Again this is just an arbitrary choice of an entry to find.

We of course want the entire matrix A^{-1}, but to do that we are trying to find how one particular entry works and then generalize it to the full matrix.

Hence why we reduced this to focusing on a single column and then reduced that to a single entry within the column.

That strategy also kind of epitomizes why we might want to use Cramer's Rule to help - as mentioned last time, that in theory can be used to find a single element of the solution without having to find all of them.

By Cramer's Rule we have

$$x_3 = \frac{\det [A_3(e_1)]}{\det A}$$

What is top determinant? Well expanding in third column, as that has first entry 1, rest 0, that will simply boil down to the matrix with that entry's row and column removed. And we do take the sign of the position of the entry into account also, so just becomes the cofactor of that entry.

Again this was for the entry in third row, first column of A^{-1}, relates to the cofactor of A in first row, third column. column. The same idea will work for every entry which will give us

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

So does that just mean replace each entry by its cofactor and divide by determinant to get inverse?

No, not quite! Each cofactor is transposed to its corresponding position, so it's transpose of the matrix of cofactors.

Fact that transpose is involved like this, related to inverse, you begin to see why transpose is important!

The matrix with these transposed cofactors, not including the determinant factor in front, is important enough that we give it a name, the <u>adjugate matrix</u> (denoted adj A). So an immediate application of that is

$$A^{-1} = \frac{\text{adj } A}{\det A}$$

Consider $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ which is known to have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Never really clarified where this came from. Ok it can be brute forced checked to work. Or found via row reducing the augmented matrix $[A\ I]$ as usual.

But the most natural way to find this from scratch is via using the adjugate formula.

I.e., cofactor of each entry is its "diagonal opposite" with an appropriate sign. Then transposing and using the known determinant gives this inverse formula.

But is this a good way in general to actually find inverse? For 3 x 3 still have to find ten determinants, not likely to be faster than the method of using augmented matrix. Maybe if many 0 entries then not so bad, e.g.

$$A = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 3 \\ 1 & 0 & 0 \end{bmatrix}$$

The cofactor of any zero entry here is 0 as when we remove the row + column through any such entry we get a 2 x 2 matrix with a zero row or column. Thus really only need three cofactors as follows:

$$C_{12} = -\begin{vmatrix} 0 & 3 \\ 1 & 0 \end{vmatrix} = 3$$

$$C_{23} = -\begin{vmatrix} 0 & 2 \\ 1 & 0 \end{vmatrix} = 2$$

$$C_{12} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6$$

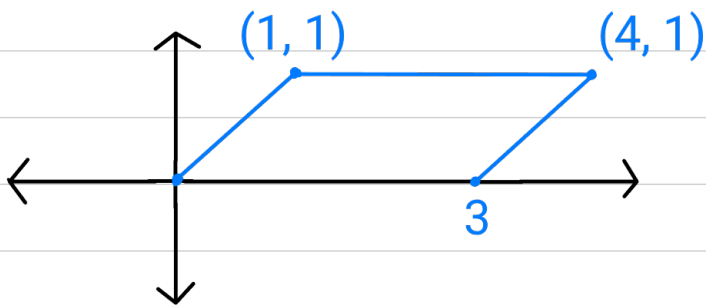
And the determinant of the matrix, expanding in first row, based on the just mentioned cofactor is also 6. Thus

$$A^{-1} = \frac{1}{6}\begin{bmatrix} 0 & 0 & 6 \\ 3 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

So bit easier than generic 3 x 3 due to all those 0 cofactors. But other method would also be proportionally easier due to that so is this really the best way? Debatable for sure.

One nice application of determinants is their relationship to areas/volumes. Not super relevant in this class, more just illustrates that determinants do have real world applications.

Consider (1, 1) and (3, 0) in $\mathbb{R}^2$



$$B = \begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix}$$

That is, we write the two vectors which generate this parallelogram as the rows of a matrix.

We might also think of these as the columns of a matrix since we are more used to that. But if we want to alter the vectors via linear combinations, i.e. row operations, more natural to think of them as rows.

Remember determinant of matrix is same as that of its transpose so that won't affect determinant.

And determinant is what we are after here, note that

$$|\det B| = 3$$

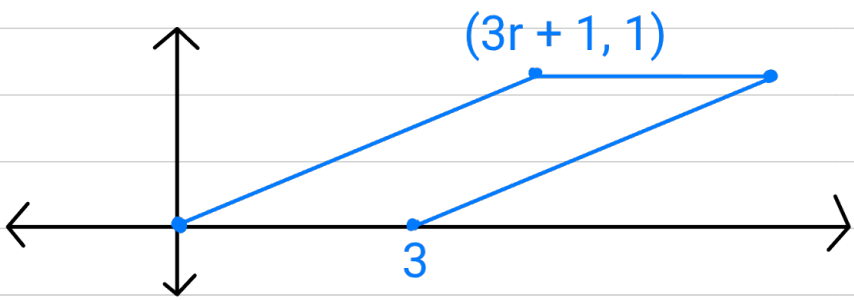
This is also the area of that parallelogram. In general $|\det|$ equals that area for any two vectors. Why?

For a 2 x 2 diagonal matrix this works since it's just a rectangle stretched/compressed. And in general any matrix can be changed to diagonal via row interchanges and adding multiple of rows to others.

I.e. we don't need scaling, that's only really necessary to make the leading entries 1 for RREF. So since the two row operations we need don't affect the absolute value of the determinant, that's largely why it works.

For instance,

$$(1, 1) + r(3, 0) = (3r + 1, 1)$$



Key is same area despite very different picture, that also helps validate why this makes sense.

More generally if you apply an invertible linear transformation on $\mathbb{R}^2$ what happens to a parallelogram? Becomes another parallelogram but how do the areas compare?

If P is the parallelogram and T the transformation, then

$$\text{Area } [T(P)] = |\det T| \cdot \text{Area } P$$

Multiplies area by absolute value of determinant. Why?

Represent P via matrix as earlier, also $T(P)$, and T itself. Then you have essentially as matrices:

$$TP = T(P)$$

So then

$$|\det T| \cdot |\det P| = |\det T(P)|$$

since determinant of product is product of determinants.

And this, since latter two are the areas, establishes the stated result.

Can also use this in three dimensions and beyond. Even if can't visualize this in higher dimensions, the idea doesn't change.

That's one essential part about all this, often the number of dimensions doesn't matter. Sure the calculation becomes more extreme but ideas same.

As alluded to, question of finding point closest to piece of space it's not in. Formula for that in three dimensions, but doesn't work in higher. Linear Algebra ideas can handle it in any dimension though.

Chapter 4: Vector Spaces

Now the content begins to get harder, more conceptual and often working in less familiar settings, e.g. spaces which are not the real numbers, definitely have to change mindset a bit.

That said, this can also be more convenient as it tends to be less calculation based. For instance Cramer's Rule which we ended last chapter with is very calculation based, not a lot of nuance though. This chapter's content is very different.

4.1 Introduction and Subspaces

A <u>vector space</u> is a set of vectors which has a defined addition of two vectors and also a defined scalar multiplication using scalars from another set (usually $\mathbb{R}$ in this class but important to note the scalars might be totally different from the vectors).

Again usually our set of scalars is $\mathbb{R}$, we say that the vector space is "over $\mathbb{R}$ " and is called a <u>real vector space</u>. We could certainly take vector spaces over $\mathbb{C}$, $\mathbb{Q}$ but not typical context in this class. The "scalars" of the vector space come from that set.

The vectors plus the two operations (addition and scalar multiplication) must satisfy many properties. to be designated a vector space. Would not be meaningful without restrictions, if it was defined "arbitrarily".

Will not list them all (are in book and very long to list all) . But will make mention of several of the more prominent ones (denote V as the vector space).

(1) Must be <u>closed</u> under both operations. That is, sum of two vectors in set must still be in set. And any scalar multiple of any vector must still be in set.

(2) Addition must be commutative. That is, $\forall$ $u, v \in V$ we have $u + v = v + u$

(3) There must be a "0" vector, i.e. a vector satisfying $u + 0 = 0 + u = u$ $\forall$ $u \in V$

(4) Each $u \in V$ has an "additive inverse", i.e. vector $-u$ for which $u + (-u) = (-u) + u = 0$

(5) Must exist a "0" scalar which satisfies $0u = 0$ $\forall$ $u \in V$

(6) Distributive laws. E.g. $c(u + v) = cu + cv$

where c scalar, u, v vectors. This is very important as it shows the two operations are connected, they can't be independent from each other in how they work.

This distributive law, like many other properties, might seem totally obvious because you are so used to using it in real numbers. But in a different setting where the operations are entirely different it might not hold - you have to check!

That is one reason why this can be hard for folks, you often have to go against your instincts in what is true. Something that is obvious in real numbers may not be so in other contexts, in fact might not be true at all, again must be careful.

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(7) Must also be a "1" scalar, i.e. a multiplicative identity which satisfies

$$1 \cdot u = u \cdot 1 = u \qquad \forall \quad u \in V$$

Is there a "1" vector? A vector which when multiplied by any other vector equals the latter vector?

There is not! Vector space doesn't inherently allow multiplication of two vectors. Sure you might define it for whatever purpose in certain situations, but it is not inherently defined in vector space in general. Only addition and scalar multiplication.

<u>Ex:</u> Say you take all continuous functions from $\mathbb{R}$ to $\mathbb{R}$. It makes sense to add two such functions (stays continuous) as does multiplying by a constant.

This is a vector space, but one which is very, very different from $\mathbb{R}^m$, after all these "vectors" are functions not points!

Again, need to practice this, hard for people to adapt such things, if you want to say prove something you likely can't use matrices as the "vectors" are just too different than the points in $\mathbb{R}^m$.

Common vector space we have worked in is $\mathbb{R}^m$ over $\mathbb{R}$. Typical question was with an $m \times n$ matrix did the columns span $\mathbb{R}^m$.

A pivot in every row was key, or if matrix was square check for invertibility.

But if columns don't span $\mathbb{R}^m$ then what do they span?

We talked about this, often referred to as some "piece" of the space. But really the proper technical term is it's a <u>subspace</u> of the original space.

<u>Subspace:</u> A subset of a vector space which is a vector space in its own right.

Many subsets are not subspaces, subspace is very special type of subset.

Be careful not to confuse those two terms (common mistake!), many folks think they mean the same thing.

As an example, in the real vector space $\mathbb{R}$ we have $\mathbb{Z}$, the set of integers, as a subset, but it is not a subspace since a real scalar multiple of an integer is not necessarily still an integer.

How to check if subset is subspace? Well there are many properties needed for vector space, do we really have to check them all? Fortunately not!

Since already contained in a vector space, some properties automatically carry over like commutativity of addition. We only have to check three things:

(a) 0 vector is in subset, i.e. $0 \in W$

(b) Closed under addition, i.e. $w_1, w_2 \in W$ implies that $w_1 + w_2 \in W$ as well

(c) Closed under scalar multiplication, i.e $w \in W$ and c any scalar in the proper set implies that $cw \in W$

One very good example are the matrix equations $Ax = 0$ and $Ax = b$ for some $b \neq 0$, i.e. homogeneous vs. nonhomogeneous.

We talked about how solution set for former is the span of a set of vectors, for latter it is not.

But again, technical way to say it is that solution set for homogeneous equation is a subspace (clearly satisfies the three properties) while 0 is not solution to nonhomogeneous equation so that solution set is clearly not a subspace.

This kind of indirectly implies that subspaces may have some type of correlation with spans of sets of vectors.

And indeed they do! That is the next big topic.